

Liam J. Nolan

1005 E Delaware Ave.
Urbana, IL, 61801

(703) 713-5872 (m)
liamjn2@illinois.edu

EDUCATION **University of Illinois Urbana-Champaign (UIUC), Dept. of Astronomy**

Graduate Student; Urbana, IL; GPA: 3.96

- Advisor: Xin Liu
- Graduate Council Outreach Chair (2022-2024)
- NSF Graduate Research Fellow (2024-current)

Arizona State University (ASU)

B.S. 2022 Summa Cum Laude; Tempe, AZ; GPA: 4.0

- Degrees: Earth & Space Exploration (Astrophysics), Physics; Minor: French
- Undergraduate Researcher and Outreach Coordinator in Rogier Windhorst's Cosmology Research Group
- Senior thesis producing SEDs of VLA-identified population of faint radio galaxies in the *JWST* North Ecliptic Pole Time-Domain Field

Thomas Jefferson High School for Science and Technology

Graduated June 2018; Alexandria, VA

Advanced Studies Diploma

- Developed decision-tree classifier for black hole identification – winner of American Geophysical Union Virtual Poster Showcase (High School Division)

Coursera

- Data-Driven Astronomy Course Certificate (2017)
- Course on utilization of Python-based data in astronomy

RESEARCH EXPERIENCE **Frequency and Abundance of Binary sUpermassive bLack holes from Optical Variability Surveys (FABULOVs): Hubble Space Telescope Imaging of Radial-Velocity Selected Binary Candidates 2025**

- Study on candidate dual sub-pc AGN in *HST* imaging, focused on host characterization with PSF/Sersic subtraction via GALFIT
- Determination of proportion of dual AGN candidates in recently merged galaxies
- DOI: [10.3847/1538-4357/ade95](https://doi.org/10.3847/1538-4357/ade95)

Center for Astrophysics | Harvard & Smithsonian

REU Intern; Cambridge MA; 6/2021 – 8/2021

- Developed n-body simulations of dynamics of Hierarchical Triple Systems with Rosanne Di Stefano

Arizona State University / NASA Space Grant

Scholar/Recipient; Tempe, AZ; 8/2019 – 5/2021

- Undergraduate Research Program with a focus on outreach, awarded twice
- Conducted original research into faint radio galaxies in the JWST North Ecliptic Pole Time Domain Field with Dr. Rolf Jansen (ASU)
- Determined low ($\sim 10\%$) proportion of AGN-dominated vs. star formation dominated galaxies in field, avoiding the need for expensive FIR data

Interactive Cosmology Visualization Using the Hubble UltraDeep Field Data in the Classroom

Principal Investigator; SESE (ASU); Tempe, AZ; 4/2020 – 12/2020

- Led study on the use of a new virtual teaching tool, Appreciating Hubble at Hyperspeed (AHaH), which allows visualization of the HUDF in 3D
- Developed study on the use of the tool in introductory astronomy classrooms, with highly positive feedback from students and instructors
- Published in the AEJ: <https://doi.org/10.32374/AEJ.2021.1.1.003>

Smithsonian National Air & Space Museum

Intern; Education Dept., CEPS; Washington, DC; 4/2019 – 7/2019

- Conducted research on TARS, a classification of aeolian features on the surface of Mars; developed an original analysis pipeline and associated documentation
- Developed educational presentations on supernova remnants and gravitational waves
- Worked with the Center for Earth and Planetary Science's Jim Zimbelman, and NASM Education Dept.

TECHNICAL SKILLS

- Programming in Python, including astronomy and data science packages: Anaconda, NumPy, SciPy, AstroPy, SciKit-Learn, Pandas, Matplotlib
 - Linux (CentOS), as well as set-up and use of Linux Virtual Machines (VirtualBox)
 - Fortran
 - IRAF, ds9, WCSTools, GALFIT, other astronomy computational tools
 - Analysis of VLA, VLBA, HST, and LBT datasets
 - Data collection and analysis from SIMBAD astrophysics data archive
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PRESENTATIONS AND TEACHING

Adler Planetarium: Astronomy Conversations

Public Presentation Series; Urbana, IL; 8/2024 – current

- Public engagement program at the Planetarium
- Organizing UIUC astronomy's participation on an approximately quarterly basis

Jump into Science

Public Presentation Series; Urbana, IL; 8/2024 – current

- Series of talks and demonstrations at Urbana Free Library for elementary-aged children
- Developed 4 presentations and interactive activities

Science at the Market

Public Presentation; Urbana, IL; 5/2024

- Part of a regular series engaging the public at local farmer's market

Terre Haute: Eclipse 2024

Public Presentation; Terre Haute, IN; 4/2024

- Public talk at Rose-Hulman University on the day of the 2024 total solar eclipse
- Culmination of ~6 months' organization of multiple events in IL and IN

Mahomet Library: Eclipse 2024

Public Talk; Urbana, IL; 4/2024

- Presentation to seniors in advance of the 2024 total solar eclipse

Girls' Astronomy Summer Camp

Educational Camp; Urbana, IL; 6/2023

- Summer camp aimed at high-school-age girls and gender minorities interested in astronomy and broader sciences
- Developed camp application materials and curricula, and led class on spectroscopy

The Literary Space Camp

Talk Series; 2/2023 – 4/2023

- Organized and gave a series of talks aimed at young children at a local bookstore, total attendance ~100

ASU Open Door

Public Presentation; Tempe, AZ; 3/2019, 3/2020, 3/2021, 4/2022

- Annual campus-wide public engagement event, assisted in 2019, '20, '21, '22 – consistent attendance in the thousands across the event when in-person, large virtual impact of online programming in 2021
- Led a group of undergraduates from research group to present our work and that of our professional colleagues to the public

JWST Launch Event

Online Presentation; 12/2021

- Online presentation leading up to the launch of *JWST*, hosted by ASU

ASU Earth and Space Exploration Day

Public Presentation; Tempe, AZ; 10/2019, 10/2020, 11/2021,

- Annual public engagement event from the School of Earth and Space Exploration, assisted in 2019, '20, '21– consistent attendance in the thousands across the event

- Led a group of undergraduates from research group to present our work and that of our professional colleagues to the public
- Included a live interactive demonstration of the AHaH software

ASU SESE Open House

Online Talk; 10/2021

- Public presentation and organization of keynote speaker on research

ASU New Discoveries Lecture Series

Online Panel Talk; 10/2021

- Organizer and panelist for discussion of JWST launch with lead scientists & students

SAO Research Symposium

Online Talk; 8/2021

- Presented work completed during REU program, moved online due to COVID-19

ASU School of Earth & Space Exploration

Teaching Assistant; Tempe, AZ; 1/2021 – 5/2021

- Assisted Professor Rogier Windhorst with AST 322: Introduction to Cosmology
- Worked on class scheduling, grading, technical support, and instructional assistance

Arizona Space Grant Symposium

Online Talk; 4/2020, 4/2021

- Presented work for each year of grant program, moved online due to COVID-19
- Followed a poster presentation to the public earlier in the year

237th Meeting of the American Astronomical Society

iPoster Presentation; 1/2021

- Presented abstract on education work with HUDF described above
- Also presented by other team members to the Astronomical Society of the Pacific

Sun Devil Generations - Virtual Earth and Space Exploration

Online Talk; 12/2020

- Invited public presentation of my and others' work in Rogier Windhorst's Cosmology Research Group, including HUDF education work for an ASU alumni network
- Followed by a live demonstration of the AHaH software

ASU Marston Theater Virtual Night Sky Presentation

Online Talk; 12/2020

- Invited talk and live demonstration on similar subjects as above for the general public

Udvar-Hazy Air & Space Museum – Part of the Smithsonian Institution

Edu. Intern; Discovery Stations Program; Chantilly, VA; 7/2017 – 9/2017

- Developed a new educational program on black holes
- Helped manage further training and organization of new volunteers over the summer
- Continuous with 4 years of volunteering with the museum

HONORS

NSF Graduate Research Fellowship (2024)

- GRFP recognizes and supports outstanding graduate students who have demonstrated the potential to be high achieving scientists and engineers, early in their careers.

Moeur Award (2022)

- Awarded by ASU to undergraduates in the highest academic standing

The College Student Leader (2021)

- Awarded by ASU College of Liberal Arts and Sciences to its highest achieving students (<0.01%)

Sun Angel Funk Award (2021)

- Awarded by ASU College of Liberal Arts and Sciences

Phi Beta Kappa (2021)

- The oldest and most prestigious academic honor society in the United States

Barry S. Goldwater Scholarship (Nominee, 2021)

- Nominated by ASU
- The most prestigious undergraduate scholarship given in the natural sciences, engineering, and mathematics

ASU The College Dean's List

- Awarded for excellent academic performance; received every semester from Fall 2018 to Fall 2021

Judith J. Radke Outstanding Junior Award (2020)

- Awarded by ASU SILC French Faculty Board

National Merit Scholar (2018)

- Awarded by National Merit Scholarship Corporation to top < 1% of students per state

AP Scholar with Distinction (2017)

- Awarded by the College Board – Score of 3 or higher on five or more AP Exams

LANGUAGES

English (Fluent reading/writing/speaking, native speaker)
French (Proficient reading/writing/speaking)